PEGASUS
Midterm Report

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Outline

- Project Outline and Goals
- Current Progress
  - Gantt Chart
- Proposed Reformer Operation
  - Schematics
  - 3-D Representation
- Reformation - Methanol Fuel
- Alternate Fuels
- Current Technology and Applications
- Future Operations

Project Goals

- Design and Construct a prototype
- Explain in detail the operation with visual aids
    - Onboard Fuel Reformer prototype
    - Schematics
    - Video clips

Project Overview

- The main purpose of this device is to take a fuel, increase that fuel's efficiency, and significantly reduce emissions
- Secondary purpose is to be able to run on a wide variety of fuels, both petroleum and non-petroleum products
- Compact and suitable for many different sized applications

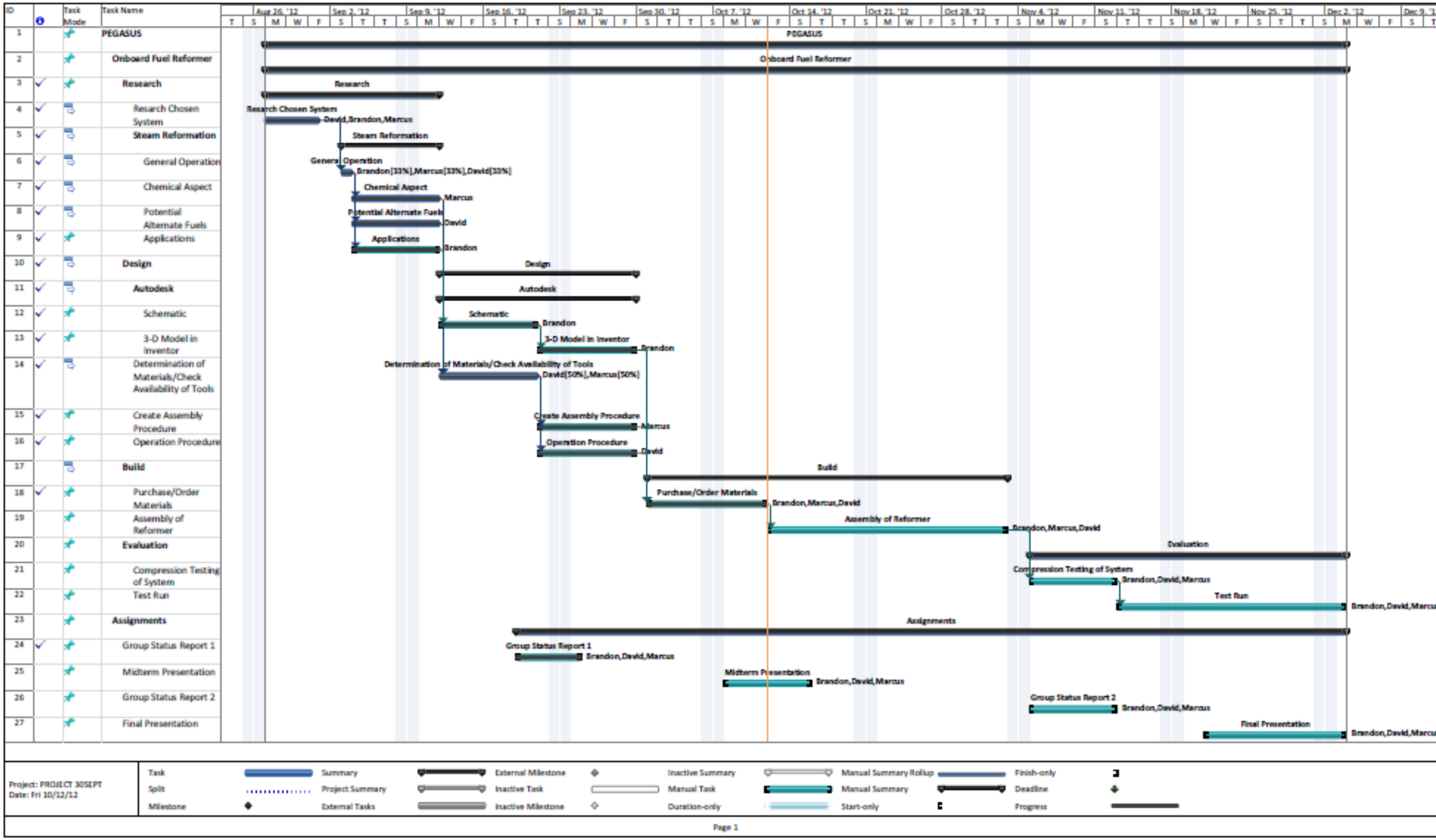
Current Tasks

- Reformer Assembly
  - Machining of parts
  - Welding/soldering joints
  - Revision of parts lists

- Updating of Assembly Procedure
  - Fine tuning procedure as changes occur

- Updating of Schematics
  - Sizes and connections may change as design is modified

GANTT Chart



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Sep 2, '12
Sep 9, '12
Sep 16, '12
Sep 23, '12
Sep 30, '12
Oct 7, '12
Oct 14, '12
Oct 21, '12
Oct 28, '12
Nov 4, '12
Nov 11, '12
Nov 18, '12
Nov 25, '12
Dec 2, '12
Dec 9, '12
1 PEGASUS
2 Onboard Fuel Reformer
3 Research
4 Resarch Chosen System
5 Steam Reformation
6 General Operation
7 Chemical Aspect
8 Potential Alternate Fuels
9 Applications
10 Design
11 Autodesk
12 Schematic
13 3-D Model in Inventor
14 Determination of Materials/Check Availability of Tools
15 Create Assembly Procedure
16 Operation Procedure
17 Build
18 Purchase/Order Materials
19 Assembly of Reformer
20 Evaluation
21 Compression Testing of System
22 Test Run
23 Assignments
24 Group Status Report 1
25 Midterm Presentation
26 Group Status Report 2
27 Final Presentation
David,Brandon,Marcus
Brandon[33%],Marcus[33%],David[33%]
Marcus
David
Brandon
David[50%],Marcus[50%]
Brandon,Marcus,David
Brandon,David,Marcus
Project: PROJECT 30SEPT
Date: Fri 10/12/12
Task
Split
Milestone
Summary
Project Summary
External Tasks
External Milestone
Inactive Task
Inactive Milestone
Inactive Summary
Manual Task
Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
Deadline
Progress
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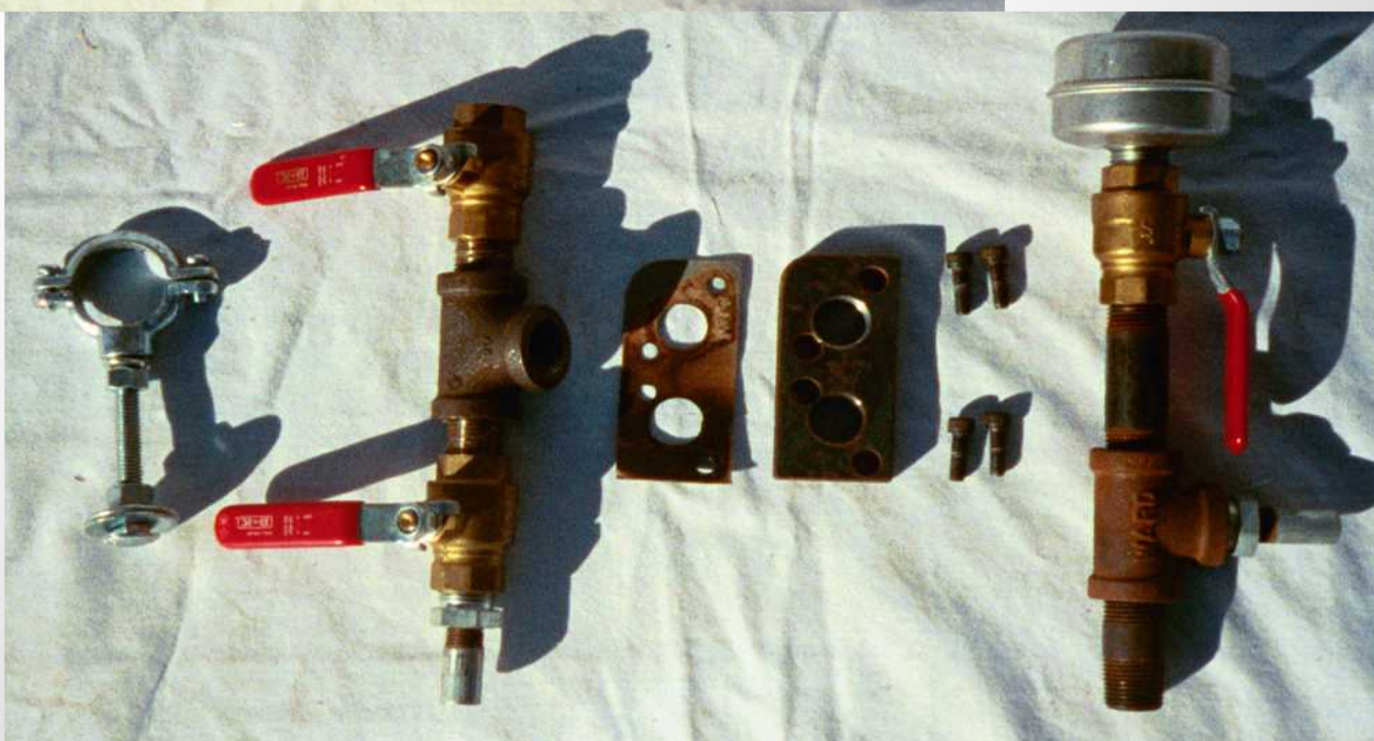
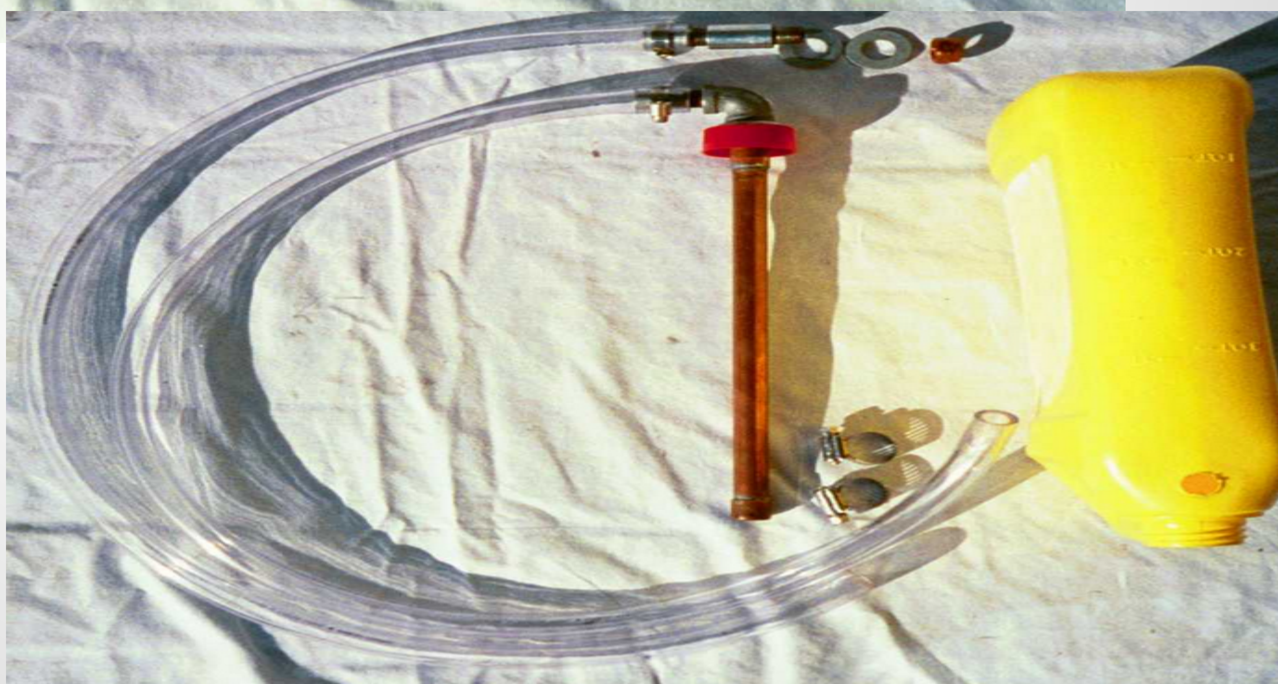
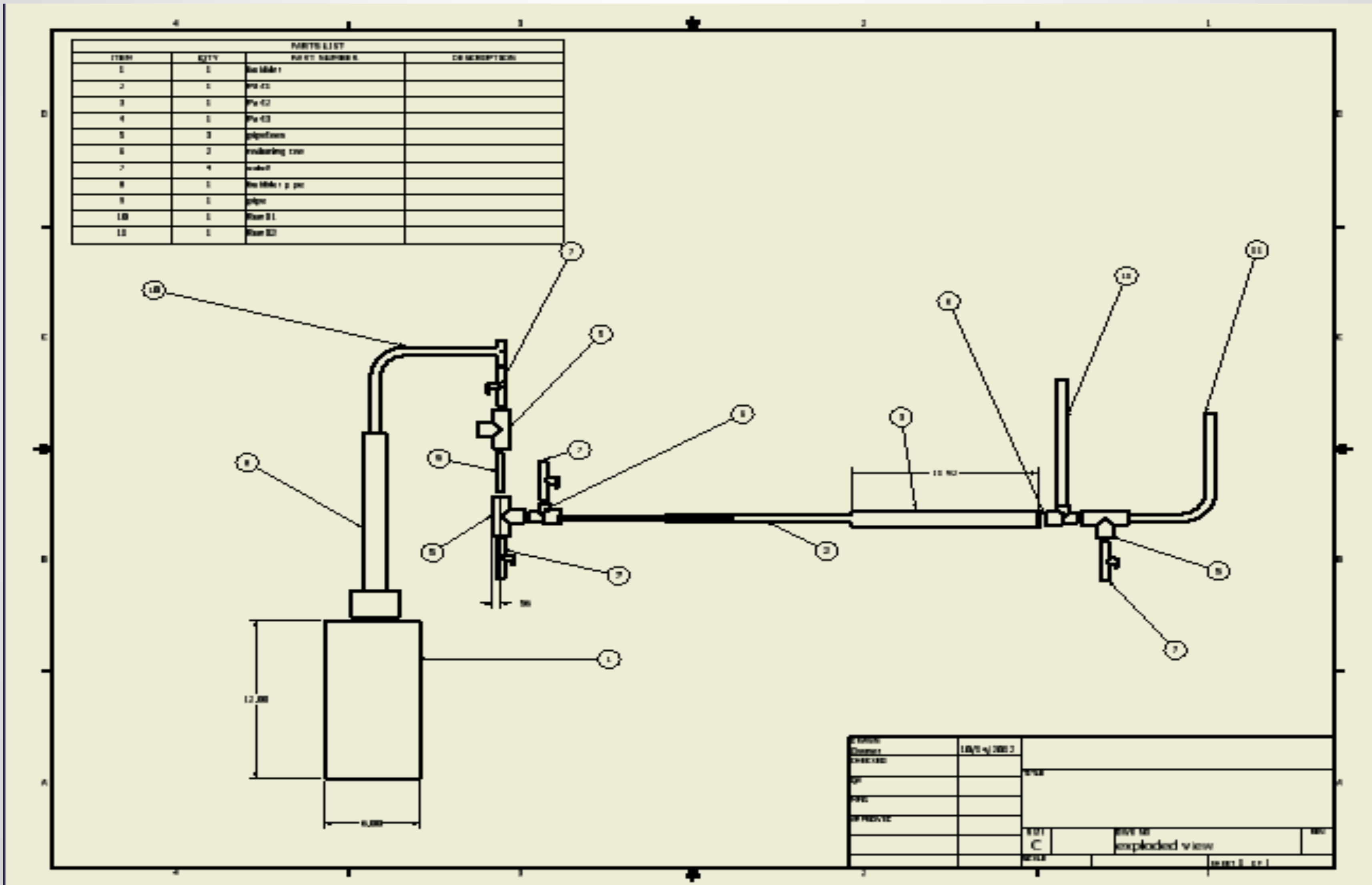
On-board Fuel Reformer

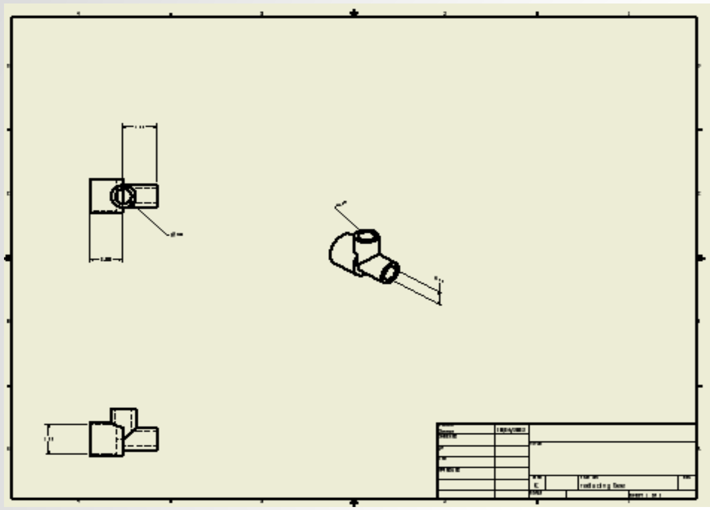


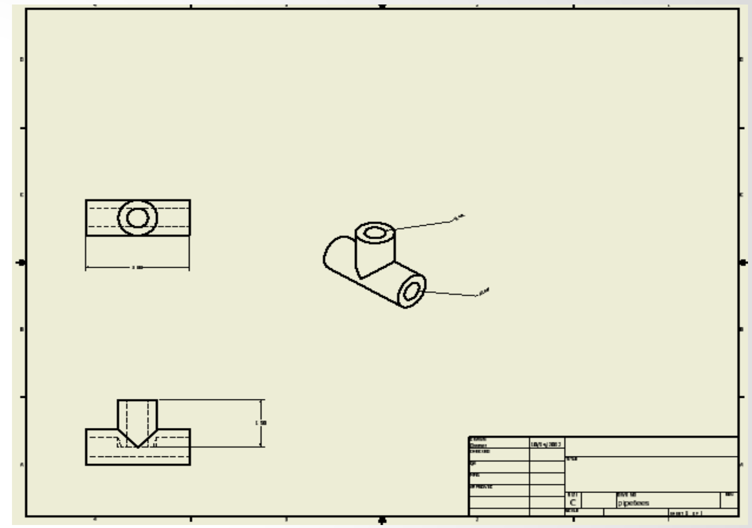


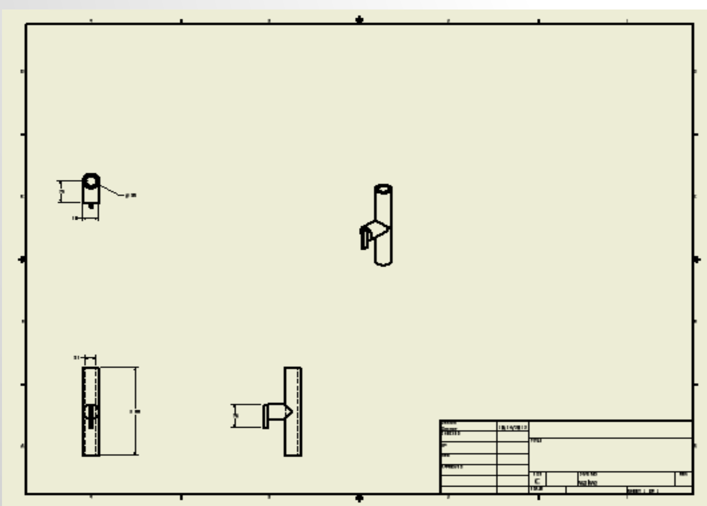
Schematics

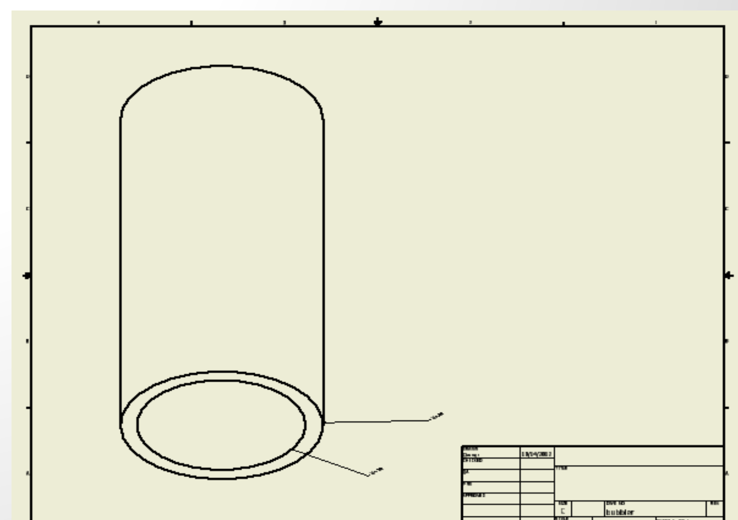


Select Parts

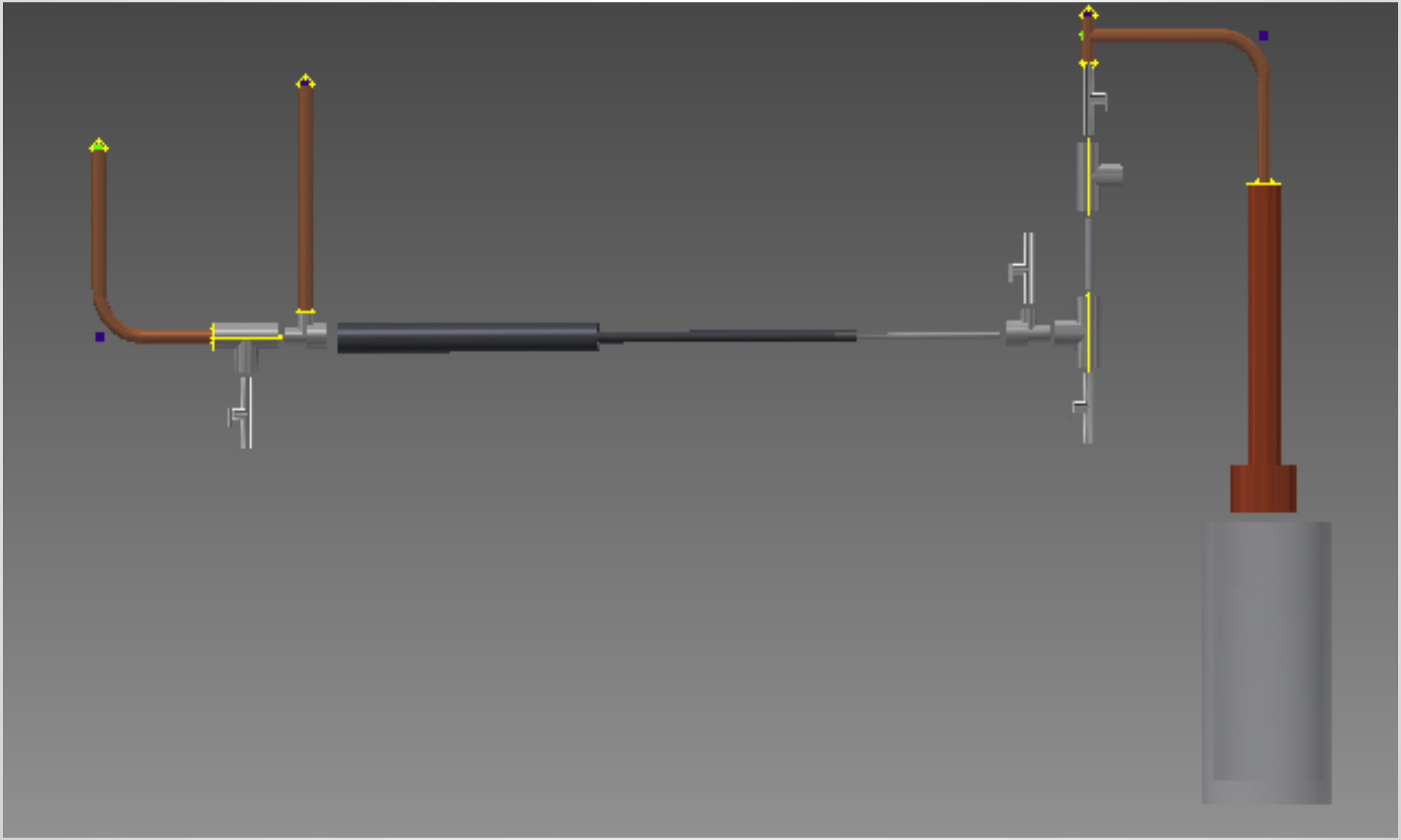








3-D Model



Chemical Aspects

- Combustion of methanol
  - $CH_3OH + 1.5(0_2 + 3.76N_2) \rightarrow CO_2 + H_2O + 5.64N_2$
- Steam Reformation
  - $CH_3OH + H_2O + HEAT \rightarrow 3H_2 + CO_2$
- Reformation using exhaust
  - $CH_3OH + 0.5(CO_2 + 2H_2O + 5.64N_2) \rightarrow 1.5CO_2 + 3H_2 + 2.82N_2$

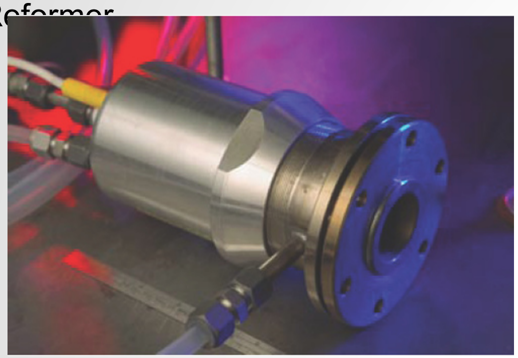
Research: Alternate Fuels

- ***Alternate fuels:***
    - Methanol
    - Ethanol
    - Liquor
    - Paint Thinner
    - Spent Motor Oil/Transmission fluid
    - Bio-fuels
        - Trans-esterification
    - Water (increases efficiency with more water added to the system, up to a threshhold)

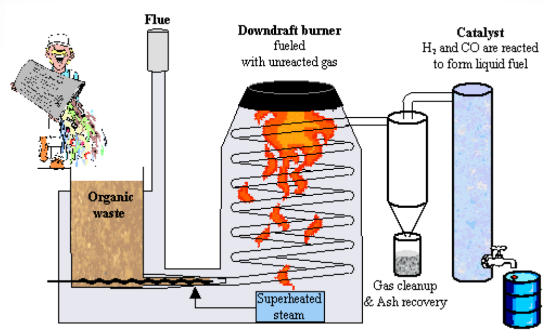
- Any liquid containing hydrogen is able to be reformed

Applications/Current Research

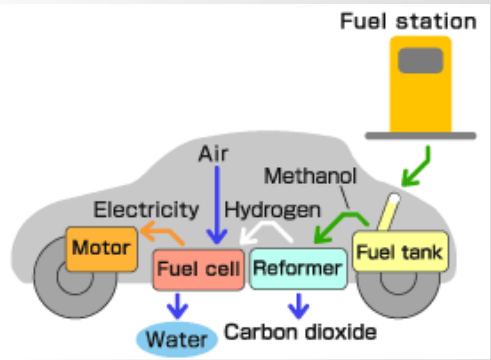
Plasmatron M.I.T.
Reformer



Organic Fuel Reformer



On-Board Methanol



Autothermal Reformer

"If widespread use of plasmatron hydrogen-enhanced gasoline engines could eventually increase the average efficiency of cars and other light-duty vehicles by 20 percent, the amount of gasoline that could be saved would be around 25 billion gallons a year," Cohn said. "That corresponds to around 70 percent of the oil that is currently imported by the United States from the Middle East."

Future Tasks through Final Presentation

- Assembly of the Reformer
- Attach Device to Lawnmower
- Establish Safety Protocol
- Begin Testing Phase with Different Fuels
- Adjustments and Modify

Questions?